

Numerical Modelling of a Sorption-Enhanced Methanation System

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SUPPLEMENTARY MATERIAL

Nomenclature

A_{ij}	Binary interaction, -
a_v	Solid specific surface area per unit volume, $m^2 \cdot m^{-3}$
c_{pg}	Heat capacity of the gas stream, $J \cdot mol^{-1} \cdot K^{-1}$
c_{ps}	Heat capacity of solid, $J \cdot mol^{-1} \cdot K^{-1}$
d_p	particle diameter, m
d_t	reactor diameter, m
D	Molecular diffusivity, $m^2 \cdot s^{-1}$
$D_{ar,i}$	effective radial dispersion coefficient of species i, $m^2 \cdot s^{-1}$
$D_{ax,i}$	effective axial dispersion coefficient of species i, $m^2 \cdot s^{-1}$
$D_{eff,i}$	Effective diffusion of species i in solid, $m^2 \cdot s^{-1}$
$D_{K,i}$	Knudsen diffusion of species i in solid, $m^2 \cdot s^{-1}$
E_A	Activation energy, $kJ \cdot mol^{-1}$
h_f	Gas-solid heat transfer coefficient, $W \cdot m^{-2} \cdot K^{-1}$
h_w	Wall heat transfer coefficient, $W \cdot m^{-2} \cdot K^{-1}$
$k_{m,i}$	gas-solid mass transfer coefficient for i-specie, $m \cdot s^{-1}$
K_{LDF}	linear driving force coefficient, s^{-1}
j_D	Chilton-Colburn factor for mass transfer, -
j_H	Chilton-Colburn factor for heat transfer, -
MW	Molecular weight
N_{WP}	Weisz-Prater number, -
P	Pressure, bar
P_i	Partial pressure, bar
P_{sat}	saturation vapor pressure of adsorbate, Pa
Pr	Prandtl number, -
q^*	equilibrium adsorption capacity, $mol \cdot kg^{-1}$
R_{gas}	Gas constant, $J \cdot mol^{-1} \cdot kg^{-1}$
Re	Reynolds number, -
r_{pore}	Pore diameter, m
Sc	Schmidt number, -
T	Temperature, K
v	Gas velocity, molecular volumes
y	Gas molar fraction, -

Greek letters

β_{20}	thermal expansion coefficient at 20°C, K ⁻¹
ε_b	bed porosity, -
ε_p	solid porosity, -
l	heat conductivity, W·m ⁻¹ ·K ⁻¹
l_{ax}	Effective axial heat conductivity, W·m ⁻¹ ·K ⁻¹
l_{ar}	Effective radial heat conductivity, W·m ⁻¹ ·K ⁻¹
u	molecular volumes, -
m	viscosity, Pa·s
ρ	density, kg·m ⁻³
t	tortuosity, -

Subscripts, superscripts and abbreviations

ads	adsorbent
g	gas phase
i,j	i,j-species index (i=CH ₄ , H ₂ O, H ₂ , CO ₂)
mix	Gas mixture
s	solid phase

S.1 Mass and thermal dispersion correlations

The average molecular diffusivity in the gas mixture has been estimated as [1]

$$D_{mix,i} = \frac{1-y_i}{\sum_{j=1, j \neq i}^{nc} \frac{y_j}{D_{i,j}}} \quad (S.1)$$

In Eq. S.1 The binary molecular diffusivities for each species are obtained by the Fuller equation (Fuller et al., 1969):

$$D_{ij} = \frac{0.0143 T^{1.75}}{P \sqrt{\frac{2000}{\frac{1}{MW_i} + \frac{1}{MW_j}}} \left[(\sum v_i)^{1/3} + (\sum v_j)^{1/3} \right]^2} \quad (S.2)$$

In Eq. S.2 ($\sum v_i$) represents the molecular volumes (Perry and Green, 2013).

The Wako correlations are used for the axial and radial dispersion coefficients (Wakao and Funazkri, 1978) in details:

$$D_{ax,i} = \frac{D_{mix,i}}{\tau} + 0.5v_0d_p \quad (S.3)$$

$$D_{r,i} = \varepsilon_b(D_{mix,i}\sqrt{\varepsilon_b} + 0.1d_p v_0) \quad (S.4)$$

The effective radial and axial thermal conductivities are calculated according to (Specchia et al., 1980):

$$\frac{\lambda_{r,i}}{\lambda_g} = \varepsilon_b + \frac{1-\varepsilon_b}{0.22 \varepsilon_b^2 + \frac{2}{3}\left(\frac{\lambda_g}{\lambda_s}\right)} + \frac{Re_p Pr_i}{8.65 \left[1 + 19.4 \left(\frac{d_p}{d_t}\right)^2\right]} \quad (S.5)$$

$$\frac{\lambda_{ax,i}}{\lambda_g} = \varepsilon_b + \frac{1-\varepsilon_b}{0.312 \varepsilon_b^{2.32} + \frac{2}{3}\left(\frac{\lambda_g}{\lambda_s}\right)} + \frac{Re_p Pr_i}{2 \left[1 + 19.4 \left(\frac{d_p}{d_t}\right)^2\right]} \quad (S.6)$$

The thermal conductivity of the mixture is calculated by the Wassiljew equation (Perry and Green, 2013):

$$\lambda_g = \sum_{i=1}^n \frac{y_i \lambda_i}{\sum_{j=1}^n y_j A_{i,j}} \quad (S.7)$$

In Eq. S.7 $A_{i,j}$ are the binary interaction parameters that are calculated using the Mason and Sexena's formulation (Perry and Green, 2013).

S.2 Mass and heat transfer coefficients

Mass and heat transfer coefficients for the gas/solid system are computed (Yagi and Kunii, 1957):

$$\frac{k_{m,i}d_p}{D_{m,i}} = j_{D,i} Re_p Sc_i^{1/3} \quad (S.8)$$

$$\frac{h_f d_p}{\lambda_g} = j_H Re_p Pr_i^{1/3} \quad (S.9)$$

Where j_D and j_H are the Chilton-Colburn factor for mass and heat transfer respectively. The wall heat transfer coefficient is as follow (Specchia et al., 1980).

$$h_w = \frac{\lambda_g}{d_p} \left(2 \varepsilon_b + \frac{1 - \varepsilon_b}{0.0024 \left(\frac{d}{d_p} \right)^{1.58} + \frac{1}{3} \left(\frac{\lambda_g}{\lambda_s} \right)} \right) + 0.0835 \frac{\lambda_g}{d_p} Re_{d_p}^{0.91} \quad (S.10)$$

The dimensionless numbers are given as:

$$Re_p = \frac{\rho_g v_0 d_p}{\mu} \quad (S.11)$$

$$Sc = \frac{\mu}{\rho_g D_{m,i}} \quad (S.12)$$

$$Pr = \frac{c_p \mu}{\lambda_{g,i}} \quad (S.13)$$

The Shomate equation and data from the NIST Chemistry WebBook (IUPAC, 2007) are used to calculate, heat capacities, enthalpy of formation, and molar entropy of all the gaseous species. The ideal gas law is applied as an equation of state.

S.3 Effective diffusivities in porous catalyst

The effective diffusivities in solid consider the true diffusion path from the molecular diffusivity and Knudsen diffusivity and the Bosanquet is used:

$$D_{eff,i} = \frac{\varepsilon_p}{\tau \left(\frac{1}{D_{K,i}} + \frac{1}{D_{mix,i}} \right)} \quad (S.14)$$

In Eq. S.14 $D_{K,i}$ represent the Knudsen diffusion coefficient calculated according to (Levenspiel, 1999):

$$D_{K,i} = 9700 r_{pore} \sqrt{\frac{T}{MW_i}} \quad (S.15)$$

In M_i is the molecular weight of i specie, while $\varepsilon_p, \tau, r_{pore}$ values are reported in Table S1

Table S1 Solid parameters (Delmelle et al., 2016; Guffanti et al., 2021; Wei et al., 2021)

ε_p	0.615
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r_{pore}, m	$1 \cdot 10^{-6}$
$S_b, m^2/g_{cat}$	685

S.4 Adsorption isotherm and reaction kinetics scheme

According to. (Schaefer and Thess, 2018) $r(T)$ temperature dependence can be expressed as:

$$\rho(T) = \frac{\rho_{20^\circ C}}{1 + \beta_{20}(T - 298.15K)} \quad (S.16)$$

In Eq. S.16 $\rho_{20^\circ C}$ (996 kg.m^{-3}) and β_{20} ($2 \cdot 10^{-3} \text{ K}^{-1}$) represent the density and the thermal expansion coefficient of adsorbate at 20°C , respectively. Saturation vapor pressure, P_{sat} , is calculated according to (Kowalska and Ambrozek, 2017):

$$\log_{10}(P_{sat}) = D + \frac{E}{T} + F \log_{10}(T) + GT + HT^2 \quad (S.17)$$

The values of the parameters in Eq. S.17 are: $D = 29.8605$, $E = -3.1522 \times 10^3 \text{ K}$, $F = -7.3037$; $G = 2.4247 \times 10^{-9} \text{ K}^{-1}$, $H = 1.8090 \times 10^{-6} \text{ K}^{-2}$ (Kowalska and Ambrozek, 2017).

In methanation reaction used the rate constant are treated as Arrhenius-type, while adsorption constants as van't Hoff-type (Koschany et al., 2016):

$$k = k_0 \exp\left(\frac{E_a}{R} \left(\frac{1}{T_{ref}} - \frac{1}{T}\right)\right) \quad (S.18)$$

$$K_x = A_x \exp\left(\frac{\Delta H_x}{R} \left(\frac{1}{T_{ref}} - \frac{1}{T}\right)\right) \quad (S.19)$$

Table S2 kinetic and adsorption constants ($T_{\text{ref}}=555 \text{ K}$) (Koschany et al., 2016)

Parameter	Value
k_0	$3.4610^{-4}, \text{mol}\cdot\text{bar}^{-1}\cdot\text{s}^{-1}\cdot\text{g}_{\text{cat}}^{-1}$
E_A	$77.5, \text{kJ}\cdot\text{mol}^{-1}$
K_{OH}	$0.5, \text{bar}^{-0.5}$
ΔH_{OH}	$22.4, \text{kJ}\cdot\text{mol}^{-1}$
K_{H_2}	$0.44, \text{bar}^{-0.5}$
ΔH_{H_2}	$-6.2, \text{kJ}\cdot\text{mol}^{-1}$
K_{mix}	$0.88, \text{bar}^{-0.5}$
ΔH_{mix}	$-10, \text{kJ}\cdot\text{mol}^{-1}$

S.5 Results

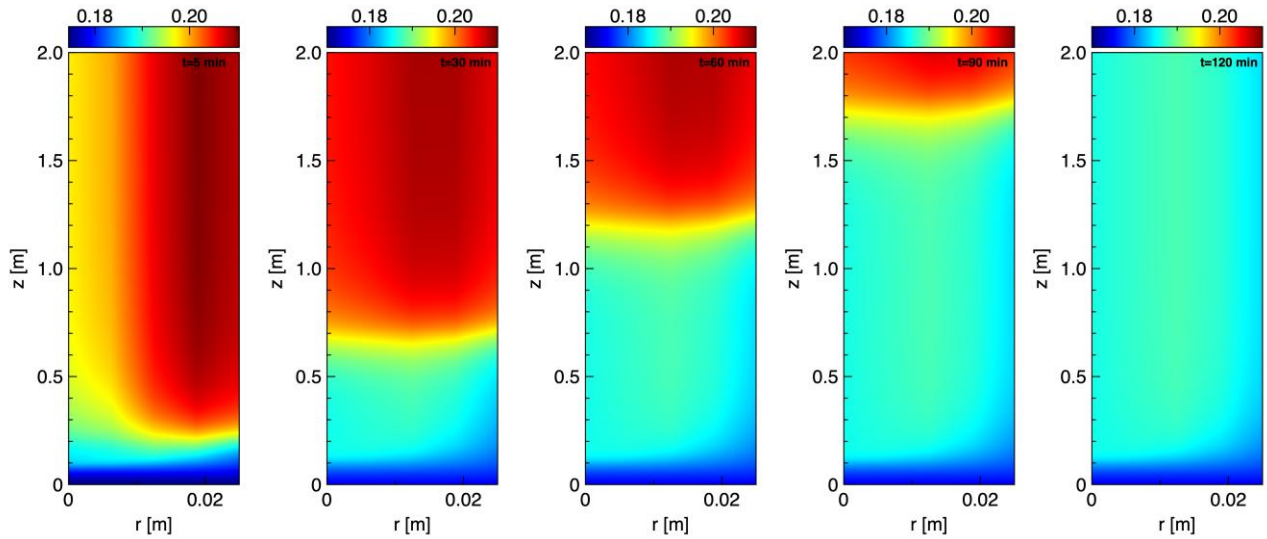


Figure S.1 superficial gas velocity (m/s) distribution at several time instants. $\text{GHSV} = 0.5 \frac{\text{m}^3}{\text{kg h}}$, $P=1$ bar and $T_{\text{in}}=300^\circ\text{C}$.

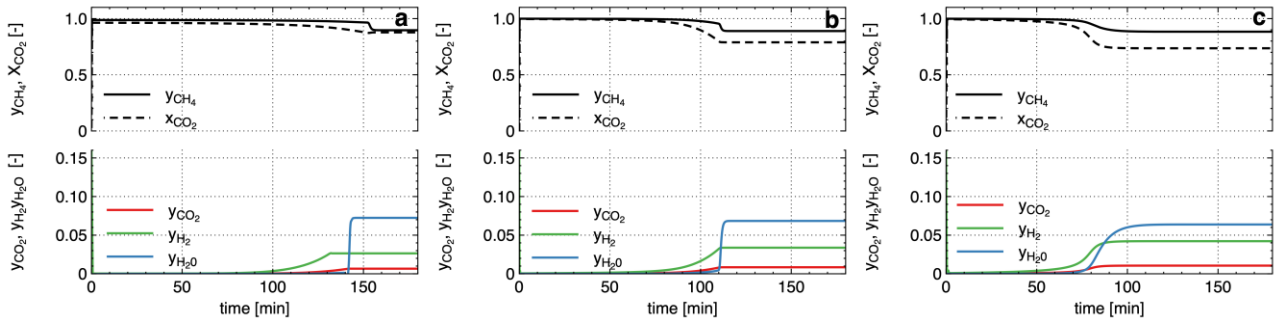


Fig. S.2: Gas composition and conversion at reactor outlet in isothermal conditions: (a) $T=280\text{ }^{\circ}\text{C}$; (b) $T=300\text{ }^{\circ}\text{C}$; and (c) $T=320\text{ }^{\circ}\text{C}$. $\text{GHSV} = 0.5 \frac{\text{m}^3}{\text{kg h}}$, $P=1\text{ bar}$.

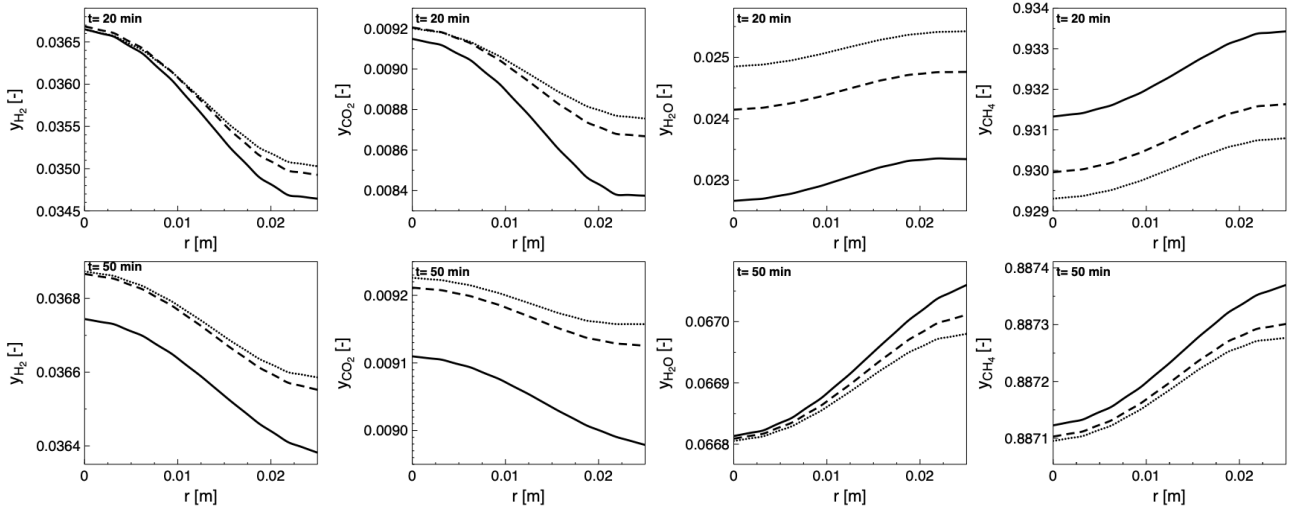


Figure S.3 Gas molar fraction profiles in the pellet at catalyst center (solid lines), surface (dashed lines) and in the gas (dotted lines) according to the legend along the traversal section at $z=0.5\text{ m}$ and for two distinct time instants. $\text{GHSV} = 0.5 \frac{\text{m}^3}{\text{kg h}}$, $P=1\text{ bar}$, and $T_{\text{in}}=300\text{ }^{\circ}\text{C}$.

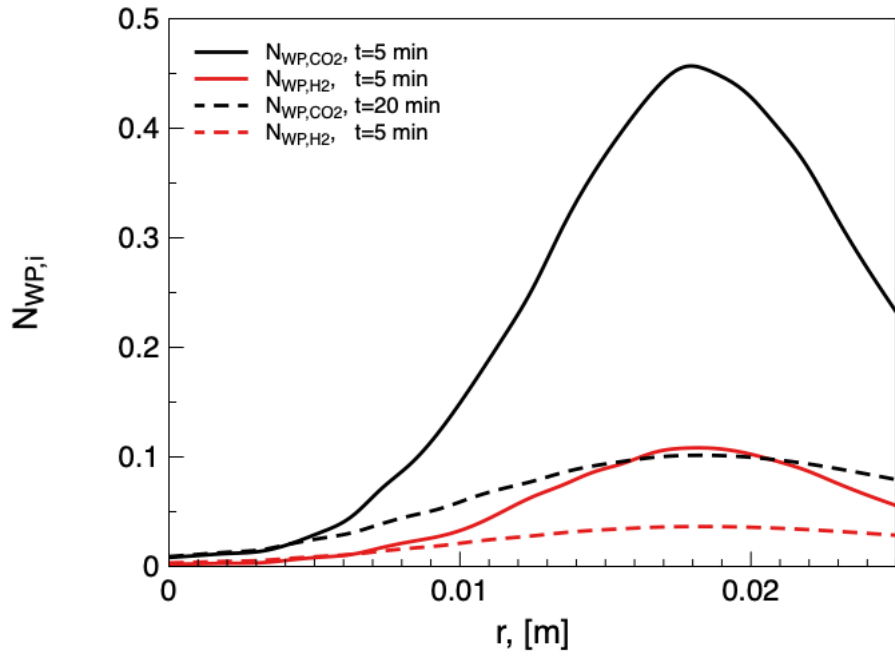


Figure S.4 Weisz-Prater number for CO₂ and H₂ specie along reactor radius at $z=0.2$ m. GHSV= $0.5 \frac{m^3}{kg\ h}$, $P=1$ bar, and $T_{in}=300^\circ C$.

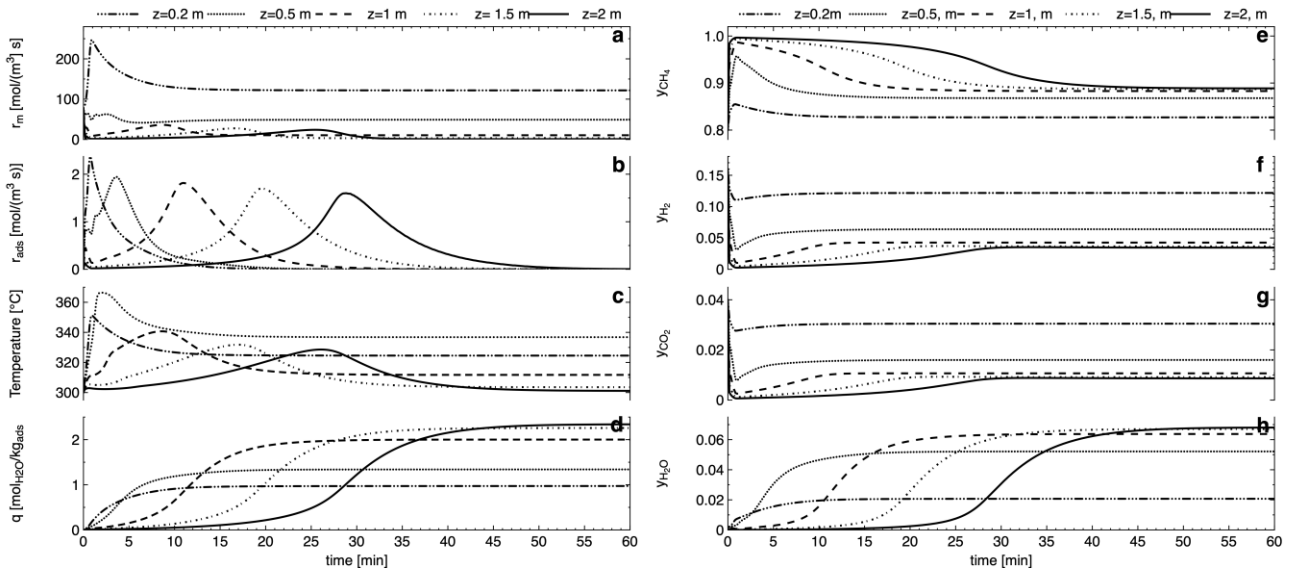


Figure S.5 (a) Average methanation reaction, (b) adsorption reaction, (c) temperature, (d) adsorbed water, and gas composition (e-h). GHSV= $1.5 \frac{m^3}{kg\ h}$, $P=1$ bar and $T_{in}=300^\circ C$.

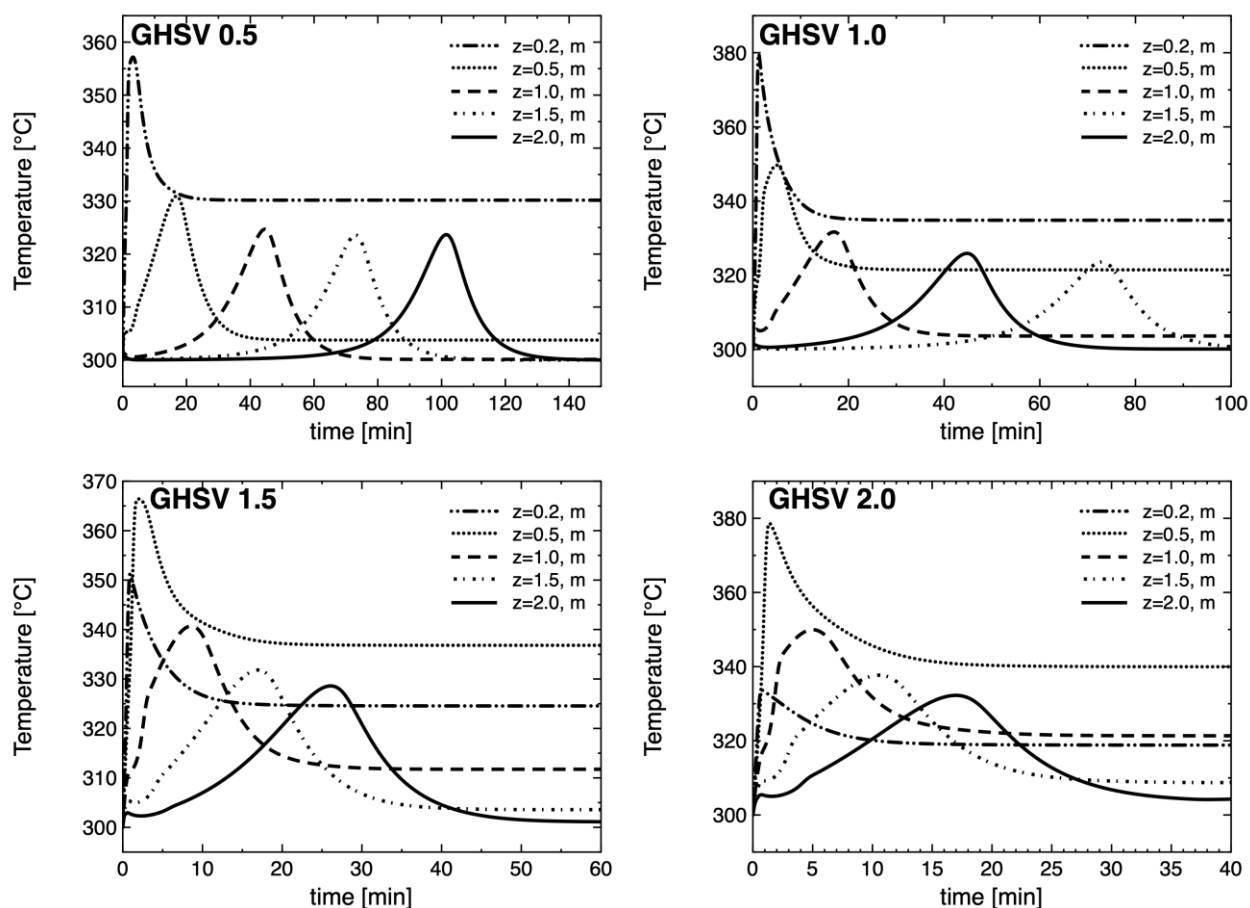


Figure S.6 Averaged Temperature on transversal section v.s. time at different reactor height for several GSHV $\left(\frac{m^3}{kg\ h}\right)$. $P=1\text{ bar}$ and $T_{in}=300^\circ\text{C}$.

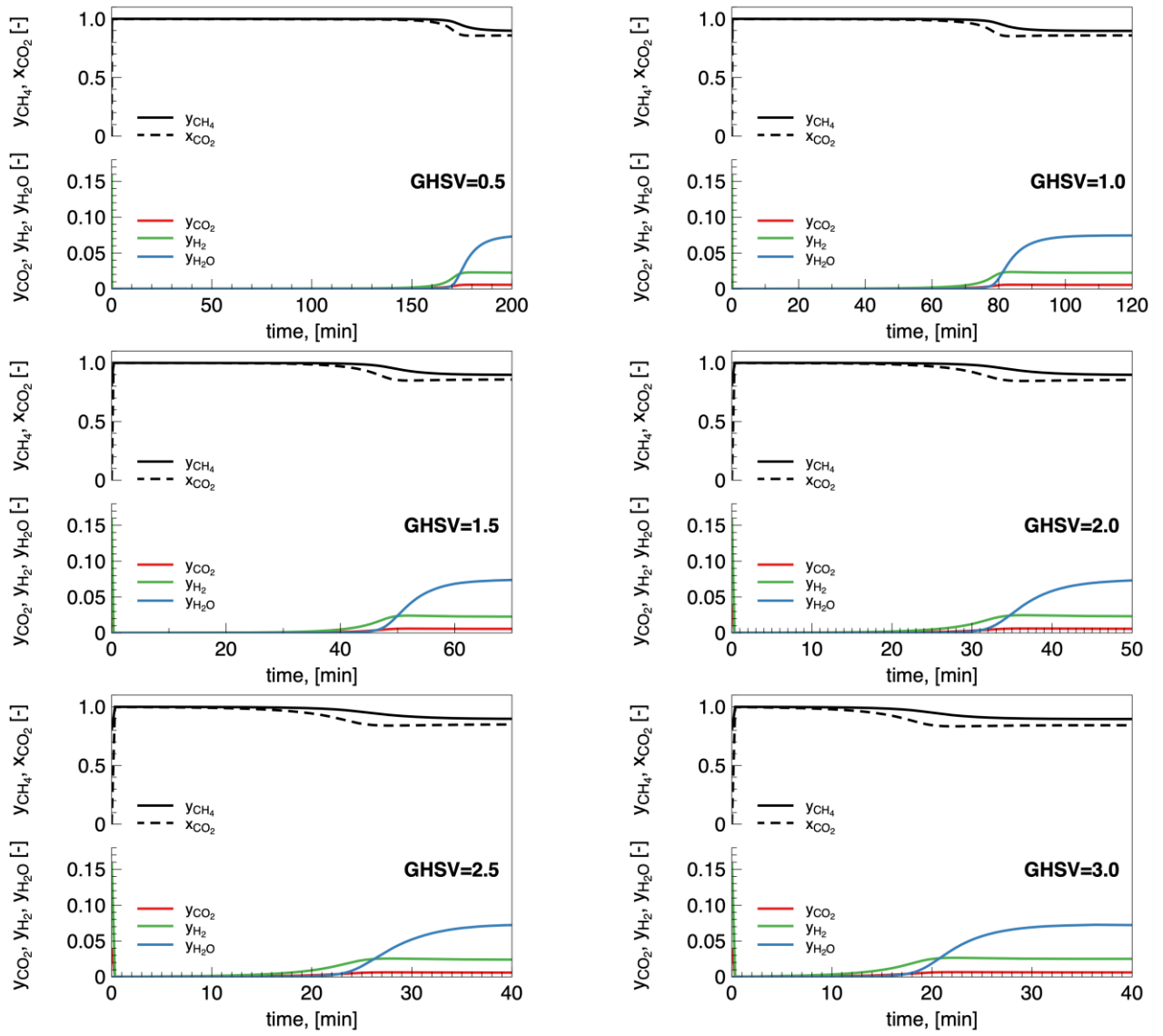


Figure S.7 Gas composition and conversion at reactor outlet for several GHSV values $\left(\frac{m^3}{kg\ h}\right)$, $P=3$ bar, and $T_{in}=300^\circ C$.

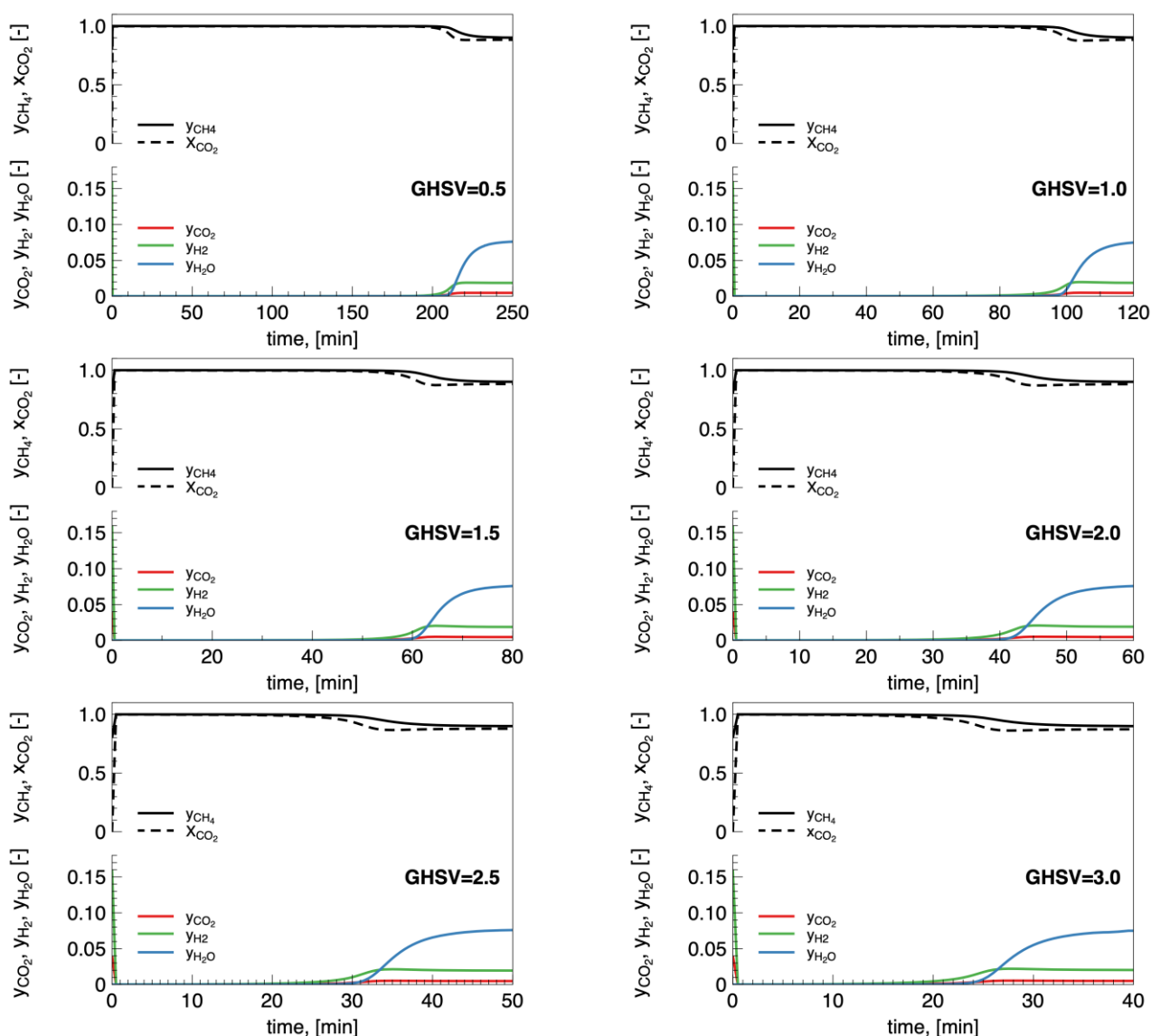


Figure S.8 Gas composition and conversion at reactor outlet for several GHSV values ($\frac{m^3}{kg\ h}$), $P=5$ bar, and $T_{in}=300^\circ C$.

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